

EXERCICE 38:

(e_1, e_2, e_3, e_4) La base canonique de $\mathbb{R}^4$
on pose

$$\xi_1 = e_1 e_4 - e_3 \text{ et } \xi_2 = e_2 + e_3$$

$$F = \text{vect}(\xi_1, \xi_2);$$

a)

$$\xi_1 = e_1 e_4 - e_3 = (0, 0, -1, 1)$$

$$\xi_2 = e_2 + e_3 = (0, 1, 1, 0)$$

ξ_1 et ξ_2 ne sont pas colinéaires $\Rightarrow$ libre

F libre et génératrice $\Rightarrow$ Base $\Rightarrow \dim(F) = 2$

b)

on cherche ξ_3, ξ_4 tq: $(\xi_1, \xi_2, \xi_3, \xi_4)$ une famille libre

~~$$\xi_1 = (0, 0, -1, 1) \quad \xi_2 = (0, 1, 1, 0) \quad \text{Soit } \alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$$~~

~~$$\xi_3 = (0, 2, 2, 0) \quad \xi_4 = (1, 0, 0, 0)$$~~

~~$$\alpha_1 \xi_1 + \alpha_2 \xi_2 + \alpha_3 \xi_3 + \alpha_4 \xi_4 = 0$$~~

~~$$\alpha_1 (0, 0, -1, 1) + \alpha_2 (0, 1, 1, 0) + \alpha_3 \xi_3 + \alpha_4 \xi_4 = 0$$~~

~~$$(0, \alpha_2, -\alpha_1, \alpha_1) + (0, \alpha_2, \alpha_2, 0) + \alpha_3 \xi_3 + \alpha_4 \xi_4 = (0, 0, 0, 0)$$~~

$$\xi_1 = (0, 0, -1, 1) \quad \text{Soit } \xi_3 = (1, 0, 0, 0)$$

$$\xi_2 = (0, 1, 1, 0) \quad \text{et } \xi_4 = (0, 1, 0, 0)$$

Soit $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$

$$\alpha_1 (0, 0, -1, 1) + \alpha_2 (0, 1, 1, 0) + \alpha_3 (1, 0, 0, 0) + \alpha_4 (0, 1, 0, 0) = 0$$

$$(0, 0, -\alpha_1, \alpha_1) + (0, \alpha_2, \alpha_2, 0) + (\alpha_3, 0, 0, 0) + (0, \alpha_4, 0, 0) = (0, 0, 0, 0)$$

$$(\alpha_3, \alpha_2 + \alpha_4, -\alpha_1 + \alpha_2, \alpha_1) = (0, 0, 0, 0)$$

$$\left. \begin{array}{l} \alpha_3 = 0 \\ \alpha_1 = 0 \\ \alpha_2 = 0 \\ \alpha_4 = 0 \end{array} \right\} \Rightarrow \text{libre et comme } \dim(\xi_1, \xi_2, \xi_3, \xi_4) = \dim(\mathbb{R}^4) = 4$$

Alors c'est une base de $\mathbb{R}^4$

Donc $\mathcal{E}_3 = e_1$ et $\mathcal{E}_4 = e_2$

$$G = \text{vect}(\mathcal{E}_3, \mathcal{E}_4) = \{(1, 0, 0, 0), (0, 1, 0, 0)\}$$

$$F = \text{vect}(\mathcal{E}_1, \mathcal{E}_2) = \{(0, 0, -1, 2), (0, 1, 1, 0)\}$$

c)

$$\text{Mq } F \cap G = \{(0, 0, 0, 0)\}$$

Soit $x \in F$ et $x \in G$ Mq $x = (0, 0, 0, 0)$

Soit $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$

$$x = \alpha_3(1, 0, 0, 0) + \alpha_4(0, 1, 0, 0)$$

$$x = \alpha_1(0, 0, -1, 2) + \alpha_2(0, 1, 1, 0)$$

$$\alpha_3(1, 0, 0, 0) + \alpha_4(0, 1, 0, 0) - \alpha_1(0, 0, -1, 2) - \alpha_2(0, 1, 1, 0) = 0_4$$

$$(\alpha_3, 0, 0, 0) + \alpha_4(0, \alpha_4, 0, 0) + (0, 0, \alpha_1, -2\alpha_1) + (0, -\alpha_2, \alpha_2, 0) = 0_4$$

$$(\alpha_3, \alpha_4 - \alpha_2, \alpha_1 - \alpha_2, -2\alpha_1) = (0, 0, 0, 0)$$

$$\alpha_3 = 0; \alpha_1 = 0; \alpha_2 = 0; \alpha_4 = 0$$

$$\text{Donc } x = (0, 0, 0, 0) \Rightarrow F \cap G = \{(0, 0, 0, 0)\}$$

d)

$$\text{Mq } F \oplus G = \mathbb{R}^4; \text{ on a } F \cap G = \{0_{\mathbb{R}^4}\}$$

$$\forall v \in \mathbb{R}^4, \exists x \in F, \exists y \in G \text{ tq } x + y = v$$

Soit $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$

$$x = \alpha_1(0, 0, -1, 2) + \alpha_2(0, 1, 1, 0)$$

$$y = \alpha_3(1, 0, 0, 0) + \alpha_4(0, 1, 0, 0)$$

$$v = \cancel{1}e_1 + \cancel{1}e_2 + e_3 + e_4$$

$$\alpha_1(0, 0, -1, 2) + \alpha_2(0, 1, 1, 0) + \alpha_3(1, 0, 0, 0) + \alpha_4(0, 1, 0, 0) = e_1 + e_2 + e_3 + e_4$$

$$(0, 0, -\alpha_1, 2\alpha_1) + \alpha_2(0, \alpha_2, \alpha_2, 0) + (\alpha_3, 0, 0, 0) + \alpha_4(0, \alpha_4, 0, 0) = (1, 2, 1, 1)$$

$$(\alpha_3, \alpha_2 + \alpha_4, -\alpha_1 + \alpha_2, 2\alpha_1) = (1, 2, 1, 1)$$

$$\alpha_3 = 1, \alpha_2 = 1, \alpha_1 = -\frac{1}{2}, \alpha_4 = -1, 5 \Rightarrow v = x + y$$

$$\alpha_1 = -\frac{1}{2}, \alpha_4 = -1, 5 \Rightarrow F \oplus G = \mathbb{R}^4$$

Q.E.D.